

AIM: Develop a JavaScript quiz app with a timer and scoring system using DOM manipulation.

Description: The JavaScript Quiz App is an interactive web application that allows users to answer multiple-choice questions within a limited time. The application uses HTML, CSS, and JavaScript to dynamically display questions, options, and results.

JavaScript DOM manipulation is used to update the question, options, timer, score, and quiz status without reloading the webpage. A countdown timer limits the time available for completing the quiz, while the scoring system calculates the user's score based on the number of correct answers. At the end of the quiz, the application displays the user's final score and provides an option to restart the quiz.

Algorithm:

1. Start the application.
2. Create a list of quiz questions with multiple-choice options and the correct answer.
3. Initialize the variables:
 - o `currentQuestion = 0`
 - o `score = 0`
 - o Set the quiz timer.
4. Use DOM manipulation to display the first question and its options.
5. Start the countdown timer.
6. Allow the user to select an option.
7. When the user submits the answer:
 - o Check whether the selected option is correct.
 - o If correct, increase the score by 1.
 - o Move to the next question.
8. Display the next question and reset the answer selection.
9. Continue the process until:
 - o All questions are answered, or
 - o The timer reaches zero.
10. Stop the timer when the quiz ends.
11. Display the final score to the user.
12. Provide a Restart Quiz option.
13. When restarted, reset the score, question number, and timer.
14. Stop.

CODE:

HTML:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>

<title>JavaScript Quiz</title>

<link rel="stylesheet" href="style.css">

</head>

<body>

<div class="quiz-container">

    <h1>JavaScript Quiz</h1>

    <div id="timer">Time: 30</div>

    <h2 id="question"></h2>

    <div id="options"></div>

    <button id="nextBtn">Next</button>

    <h2 id="score"></h2>

</div>

<script src="script.js"></script>

</body>

</html>
```

CSS-Code:

```
body {

    font-family: Arial;

    background: #f2f2f2;

    text-align: center;

}

.quiz-container {
```

```
width: 500px;  
margin: 100px auto;  
padding: 30px;  
background: white;  
border-radius: 10px;  
box-shadow: 0 0 10px gray;  
}
```

```
#options button {  
    display: block;  
    width: 100%;  
    padding: 12px;  
    margin: 10px 0;  
    cursor: pointer;  
}
```

```
#nextBtn {  
    padding: 10px 25px;  
    background: black;  
    color: white;  
    border: none;  
    cursor: pointer;  
}
```

Javascript-Code:

```
const questions = [
```

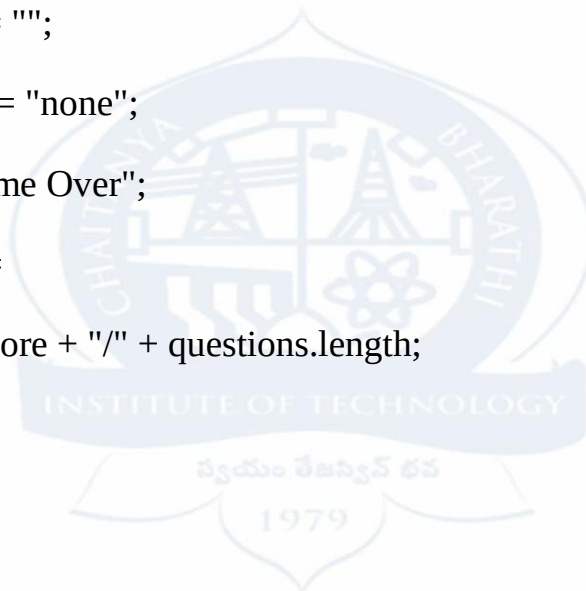


```
{
  q: "Which language makes webpages interactive?",
  o: ["HTML", "CSS", "JavaScript", "SQL"],
  a: 2
},
{
  q: "Which method selects an element by ID?",
  o: ["getElementById()", "query()", "select()", "getElement()"],
  a: 0
},
{
  q: "Which keyword declares a variable?",
  o: ["int", "var", "string", "define"],
  a: 1
}
];

let index = 0;
let score = 0;
let time = 30;
const question = document.getElementById("question");
const options = document.getElementById("options");
const timer = document.getElementById("timer");
```

```
const nextBtn = document.getElementById("nextBtn");
const scoreText = document.getElementById("score");
function showQuestion() {
  question.innerText = questions[index].q;
  options.innerHTML = "";
  questions[index].o.forEach((option, i) => {
    let btn = document.createElement("button");
    btn.innerText = option;
    btn.onclick = () => {
      if (i == questions[index].a)
        score++;
      index++;
      if (index < questions.length)
        showQuestion();
      else
        endQuiz();
    };
    options.appendChild(btn);
  });
}
let countdown = setInterval(() => {
  time--;
  timer.innerText = "Time: " + time;
```

```
if (time == 0)
    endQuiz();
}, 1000);
function endQuiz() {
    clearInterval(countdown);
    question.innerText = "Quiz Completed!";
    options.innerHTML = "";
    nextBtn.style.display = "none";
    timer.innerText = "Time Over";
    scoreText.innerText =
        "Your Score: " + score + "/" + questions.length;
}
showQuestion();
```



Laboratory Record

of _____

Roll No.: _____

Experiment No.: _____

Sheet No.: _____

Date.: _____

OUTPUT:

JavaScript Quiz

Time: 9

Which language makes webpages interactive?

HTML

CSS

JavaScript

SQL

Next



JavaScript Quiz

Time Over

Quiz Completed!

Your Score: 3/3